

AWS 3-Tier VPC Architecture – Documentation

Date 14-02-2026

Summary

In this project, I designed and implemented a 3-tier architecture in AWS to create a secure and scalable network structure. First, I created a Virtual Private Cloud (VPC) with the CIDR block 10.0.0.0/16 in the us-east-1 region. Inside the VPC, I created six subnets divided across two Availability Zones to ensure high availability. The architecture includes two public subnets for the Web tier, two private subnets for the Application tier, and two private subnets for the Database tier.

Next, I created and attached an Internet Gateway (IGW) to the VPC to allow internet access for the public subnets. I configured a public route table with a route (0.0.0.0/0 → IGW) and associated it with the web-tier subnets. To enable secure outbound internet access for the private application subnets, I created a NAT Gateway inside one of the public subnets and attached an Elastic IP to it. Then, I created a separate route table for the application tier and added a route (0.0.0.0/0 → NAT Gateway), associating it with the app-tier subnets.

For security purposes, I created a dedicated route table for the database tier with only the local route and no internet access. This ensures the database layer remains isolated and protected from direct external traffic. By separating the web, application, and database layers using different subnets and route tables, the architecture follows best practices for security, scalability, and availability in AWS.

Objective

Design a secure and scalable 3-tier architecture in AWS using:

- Public Web Tier
- Private App Tier
- Private DB Tier

VPC Configuration

- VPC Name: 3tier-vpc-test
- CIDR Block: 10.0.0.0/16
- Region: us-east-1

Subnet Configuration (6 Subnets)

Web Tier (Public Subnets)

- web-tier-1 → 10.0.1.0/24 (AZ-a)
- web-tier-2 → 10.0.2.0/24 (AZ-b)
- Auto-assign Public IP → Enabled

App Tier (Private Subnets)

- app-tier-1 → 10.0.3.0/24
- app-tier-2 → 10.0.4.0/24

DB Tier (Private Subnets)

- db-tier-1 → 10.0.5.0/24
- db-tier-2 → 10.0.6.0/24

Internet Gateway (IGW)

- Created 1 Internet Gateway
- Attached to VPC
- Used for public internet access

NAT Gateway

- Created in web-tier-1 (public subnet)
- Elastic IP attached
- Allows private app subnets to access internet securely

Route Tables (3 Total)

Public Route Table

- 0.0.0.0/0 → Internet Gateway
- Associated:
 - web-tier-1
 - web-tier-2

App Route Table

- 0.0.0.0/0 → NAT Gateway
- Associated:
 - app-tier-1
 - app-tier-2

DB Route Table

- Only local route (10.0.0.0/16 → local)
- No internet access
- Associated:

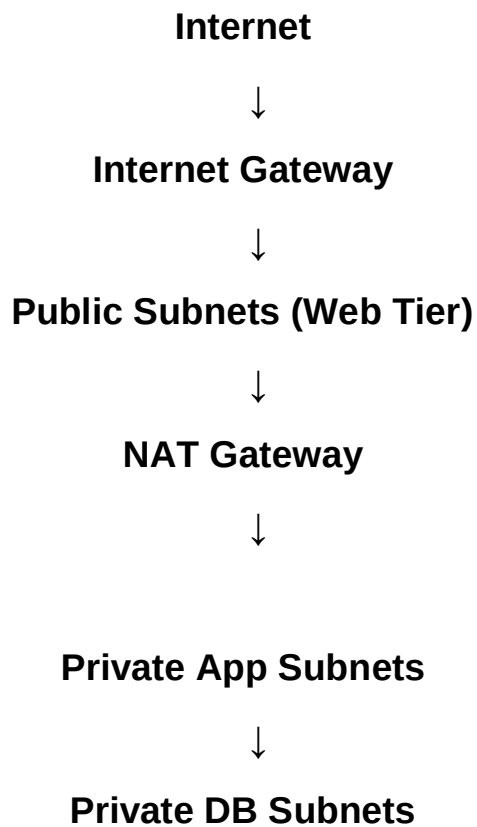
- db-tier-1
- db-tier-2

Security Design Logic

- Web Tier → Public (Internet accessible)
- App Tier → Private (Access via NAT only)
- DB Tier → Isolated (No internet route)
- Traffic flow:

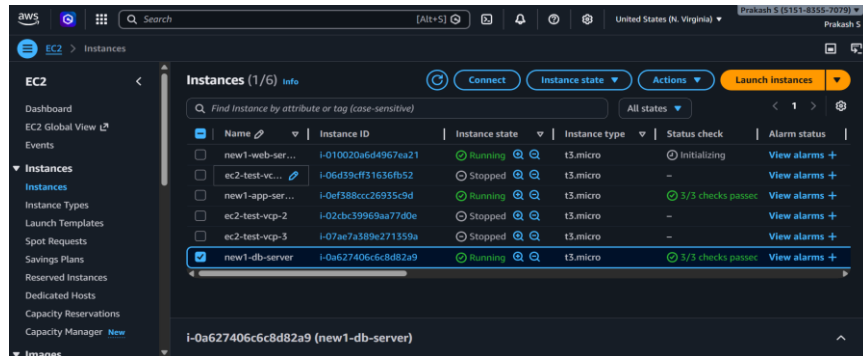
Flow: Internet → Web → App → Database

Final Architecture Flow

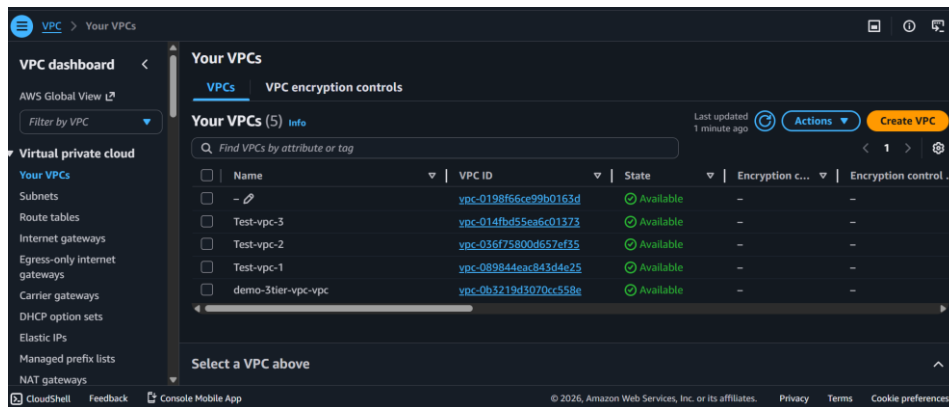


Screen Shots: VPC 3 Tier Architecture

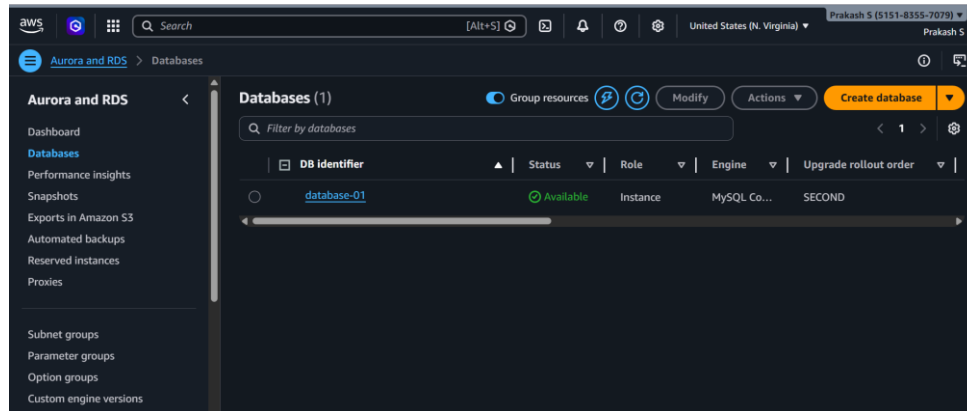
EC2 INSTANCE



AWS VPC

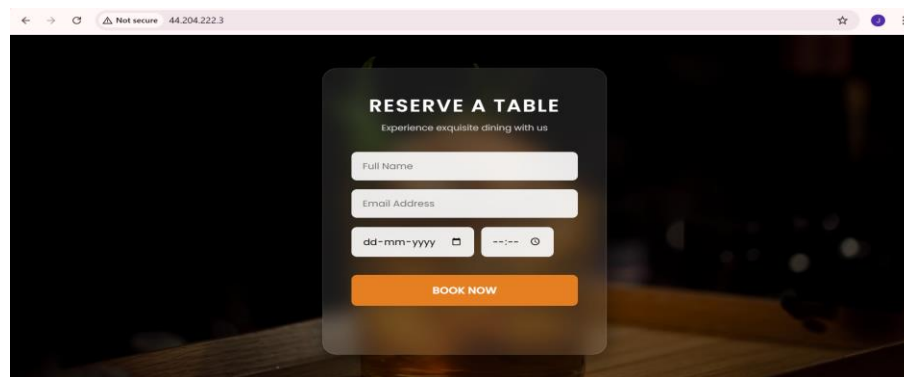


AWS RDS



DEMO : VPC 3 TIER ARCHITECTURE

Frontend



BACKEND

```

ast login: Fri Feb 13 05:13:35 2026 from 183.83.216.217
ubuntu@ip-10-0-130-126:~$ ssh -i newlweb-key.pem ubuntu@10.0.130.126
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Feb 14 06:49:26 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   42.8% of 6.7GB Processes:      109
Memory usage: 23%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.130.126

 * Ubuntu Pro delivers the most comprehensive open source security and
 * compliance features.
 * https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
0 of these updates is a standard security update.
0 see these additional updates run: apt list --upgradable
enable ESM Apps to receive additional future security updates.
see https://ubuntu.com/esm or run: sudo pro status

ast login: Fri Feb 13 05:01:16 2026 from 10.0.4.19
ubuntu@ip-10-0-130-126:~$ ls
app backend newlweb-key.pem venv
ubuntu@ip-10-0-130-126:~$ cd backend
ubuntu@ip-10-0-130-126:~/backend$ source /venv/activates
bash: /venv/activates: No such file or directory
ubuntu@ip-10-0-130-126:~/backend$ source /venv/bin/activate
bash: /venv/bin/activate: No such file or directory
ubuntu@ip-10-0-130-126:~/backend$ source venv/bin/activate
(venv) ubuntu@ip-10-0-130-126:~/backend$ python app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://10.0.130.126:5000

```

DATABASE

```
Last login: Sat Feb 14 06:49:27 2026 from 10.0.4.19
ubuntu@ip-10-0-130-126:~$ mysql -h database-01.c4x08m0kks33.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 4573
Server version: 8.0.47 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE reservation;
Reading table information for completion of table and column names
You can turn off this feature with -A

Database changed
mysql> SELECT * FROM bookings;
```

id	name	email	booking_date	booking_time	created_at
1	prakash	jothiprakash17042001@gmail.com	2026-02-05	11:11	2026-02-12 10:57:32
2	Sabari G	sabari2003@gmail.com	2026-02-13	12:00	2026-02-12 11:00:59
3	Prakash shanmugam	kalsprakash2001@gmail.com	2026-02-02	10:00	2026-02-12 11:03:26
4	surendar	jothiprakash17042001@gmail.com	2026-02-12	12:00	2026-02-12 11:05:43
5	Nickshan	Nickshan2001@gmail.com	2027-10-12	10:00	2026-02-12 12:37:01
6	Ravi	Ravi1997@gmail.com	2026-02-11	10:00	2026-02-12 12:44:16
7	vijaya	jothiprakash17042001@gmail.com	2026-02-17	16:44	2026-02-12 13:14:42
8	jj	jothiprakash17042001@gmail.com	2026-02-20	11:01	2026-02-12 13:16:49
9	AKASH kumar	Akashkumar2001@gmail.com	2026-02-16	11:00	2026-02-13 04:40:08

```
9 rows in set (0.00 sec)

mysql>
```